

Research Questions: Evidence Summary

This document consolidates all empirical evidence from the papers supporting each research question.

RQ1 — Exists

Does diagonal dominance emerge from training rather than initialization or shape compatibility?

Narrative: Why the Fingerprint Exists

Residual blocks of the form $x' = x + W_{\text{out}} \cdot \phi(W_{\text{in}} \cdot x)$ appear in every modern deep network we tested. After training, the inner product $M = W_{\text{out}} \cdot W_{\text{in}}$ develops a peculiar shape: it acts like a (typically negative) scalar multiple of the identity plus low-amplitude noise. The diagonal-dominance score $s = |\text{tr}(M)| / \|M\|_F$ captures this property in a single scale-invariant number. At initialization $s \approx 1/\sqrt{d}$ (random baseline ≈ 0.03 – 0.08); after training s rises by 1–2 orders of magnitude (≈ 4 for MLPs and GPT-2-small, ≈ 8 for GPT-2-XL). This is the fingerprint.

The mechanism is gradient-descent coupling, not dynamical isometry. Earlier work attributed the emergence of structured Jacobians in residual networks to a drift toward orthogonality. Our normalized metric $\delta_J^{\text{norm}} = \|J^T J\|_F / \|J\|_F^2 - \|d\|_F$ shows the opposite for trained transformers: pretrained GPT-2 has $\delta_J^{\text{norm}} = 0.297$ while its random-init counterpart sits at 0.025 — training drives the Jacobian away from uniform-spectrum, not toward it. The fingerprint must therefore come from somewhere other than spectral conditioning.

Three converging lines of evidence point to gradient coupling between W_{in} and W_{out} :

Tier 1 — Necessity. Shuffling W_{out} across blocks before each optimizer step (Section 9 Part B) cuts the fingerprint by 68% ($s: 3.90 \rightarrow 1.24$) while preserving task accuracy (47% vs 54%). Independence of W_{in} and W_{out} gradient updates breaks the fingerprint without breaking learning.

Tier 2 — Sufficiency. Synthetic updates that add diagonal structure directly to M , with no backprop, build the fingerprint from scratch ($s: 0.08 \rightarrow 7.57$ over 3000 steps; Section 9 Part C). The fingerprint does not require gradients of any specific shape — only correlated, persistent updates to both projections.

Tier 3 — Independence from task content. Training on shuffled CIFAR-10 labels (no learnable structure, $\sim 10\%$ test accuracy) produces a *stronger* fingerprint than normal training ($s = 5.88$ vs 3.42). Gradient flow alone, not task learning, is what creates the signature. The fingerprint is also uncorrelated with cumulative weight change (Pearson $r = -0.002$).

The fingerprint requires the residual connection. A non-residual PlainNet matched in depth and parameter count to a ResNet achieves comparable loss but shows no fingerprint ($s \approx$ chance, pair accuracy 3%; Section 12). Skip connections force every block to be a small perturbation of identity, which is what makes the diagonal structure visible — without them, blocks span the full output space and the M signature gets washed out.

Across model sizes, the fingerprint becomes more orthogonal-leaning. Mean δ_J^{norm} decreases from 0.297 (GPT-2-small) to 0.122 (GPT-2-XL), even though random-init δ_J^{norm} is essentially constant at 0.025 across the same sizes. Trained larger models converge toward — but never reach — random-init uniformity. The within-scale “training drives away from orthogonal” finding (12× gap at $d=768$) coexists with an across-scale “larger trained models are closer to orthogonal” finding (5× gap at $d=1600$). Both are real; they live on different axes.

Remaining gaps: We have not run the full optimizer sweep (SGD vs Adam vs RMSprop) that would test whether the gradient-coupling mechanism is optimizer-specific. We also have no learning-rate or batch-size ablation. These are queued under Pending Experiments.

Core Finding

The diagonal-dominance fingerprint is a **learned property** induced by gradient descent, not an artifact of initialization, matrix shapes, or residual architecture alone.

Key Evidence

1. Null Model Prediction For randomly initialized weights: - All pairs satisfy $E[s(i,j)] = 1/\sqrt{d}$ (random baseline) - Hungarian matching recovers correct pairs at chance rate $1/L$ - Verified across 7 initialization schemes

What this rules out: 1. Diagonal dominance is NOT an artifact of compatible matrix shapes 2. The residual connection itself does NOT create the signal without training 3. Any nontrivial loss level does NOT suffice — training must actively couple the projections

2. Random Labels Experiment (Gradient Flow vs Task Learning) **Question:** Does the fingerprint require learning a meaningful task, or does gradient flow through residual connections alone suffice?

Setup: We train a 48-block residual MLP ($d=128$, $\text{hidden}=256$) on CIFAR-10 under two conditions: - **Normal labels:** Standard classification training - **Shuffled labels:** Training labels randomly permuted (fixed per seed)

Both conditions use identical architecture, optimizer (Adam, $\text{lr}=1\text{e-}3$), and training duration (100 epochs). We measure diagonal dominance metrics at epochs $\{0, 5, 10, 50, 100\}$.

Why this design: - 48 blocks matches paper’s primary examples for consistency - CIFAR-10 provides a real task (not synthetic) to test practical relevance - Shuffled labels isolate gradient flow from task structure — networks can memorize random labels (driving gradients through all layers) but cannot generalize (test accuracy = chance) - If diagonal dominance emerges in both conditions, it indicates that gradient-based weight updates — not task-relevant learning — create the signal

Results (epoch 100, 3 seeds):

Condition	Test Acc	Mean $s(i,i)$	Frac Neg Trace
Normal	$52.9\% \pm 1.2\%$	3.42 ± 0.06	100%
Shuffled	$9.9\% \pm 0.5\%$	5.88 ± 0.01	100%

Trajectory (mean across 3 seeds):

Epoch	Normal $s(i,i)$	Shuffled $s(i,i)$	Normal Neg Trace	Shuffled Neg Trace
0	0.077	0.077	50%	50%
5	4.20	5.26	100%	100%
10	4.44	5.77	100%	100%
50	3.91	6.37	100%	100%
100	3.42	5.88	100%	100%

Key Finding: Shuffled labels produce a **stronger** fingerprint ($s=5.88\pm0.01$) than normal labels ($s=3.42\pm0.06$) while completely failing to generalize ($\sim 10\%$ test accuracy = chance). This demonstrates that **gradient-based weight updates — not task-relevant learning — cause diagonal dominance to emerge**.

2b. Weight Magnitude Control (P0.5 Reviewer Concern) **Concern:** A reviewer asked whether the stronger fingerprint with shuffled labels could simply be explained by “more gradient flow = larger weight changes.” If fingerprint strength correlates with $\|W_{\text{final}} - W_{\text{init}}\|_F$, then it measures cumulative gradient magnitude rather than learned structure.

Control: We tracked total weight change $\|\Delta W\|_F$ per block throughout training.

Results (mean across 3 seeds):

Epoch	Normal s	Shuffled s	Normal $\ \Delta W\ _F$	Shuffled $\ \Delta W\ _F$
5	4.20	5.26	7.32	6.40
10	4.44	5.77	10.21	7.05
50	3.91	6.37	25.00	28.18
100	3.42	5.88	34.11	36.52

Correlation: $r(s, \|\Delta W\|_F) = -0.002$ (n=24 samples across conditions/seeds/epochs)

Interpretation: The correlation is essentially **zero**. At early epochs (5, 10), shuffled labels have **higher fingerprint strength but lower weight change** — the opposite of what the “more gradient flow” hypothesis predicts. By epoch 100, weight changes are similar but fingerprint strength differs by 72% (5.88 vs 3.42).

This rules out the trivial explanation. The fingerprint measures **structural properties of the learned weights**, not cumulative gradient magnitude.

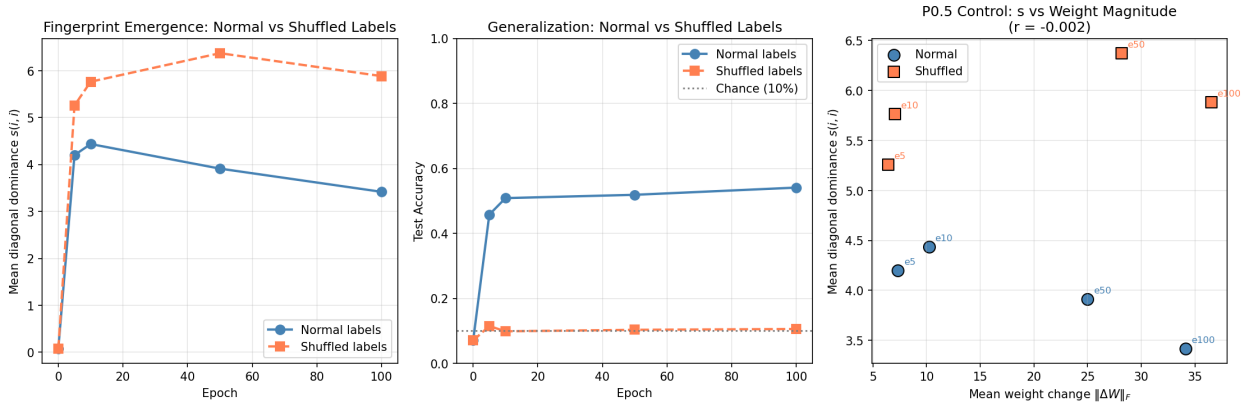


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6. Jacobian Structure Analysis (MLP) **Question:** How does the block Jacobian $J = I + M$ evolve during training, and does it relate to diagonal dominance?

Background: For residual block $f(x) = x + W_{\text{out}} \cdot \text{ReLU}(W_{\text{in}} \cdot x)$, the Jacobian is $J = I + M$ where $M = W_{\text{out}} \cdot W_{\text{in}}$. The diagonal dominance score $s = \text{tr}(M) / \|M\|_F$ measures how “scalar-identity-like” M is. We measure two orthogonality metrics:

$$\delta_J = \frac{\|J^T J - I\|_F}{\sqrt{d}}$$

(absolute deviation — sensitive to weight scale)

$$\delta_J^{\text{norm}} = \left\| \frac{J^T J}{\|J\|_F^2} - \frac{I}{d} \right\|_F$$

(normalized — scale-invariant, measures singular value uniformity)

Setup: Using a 48-block MLP ($d=128$, $\text{hidden}=256$), we compute (s , δ_J , δ_J^{norm}) for each block: - At initialization (7 schemes, 3 seeds each = 144 blocks per scheme) - After training (epochs 0, 5, 300 with kaiming_normal init, 3 seeds)

Note: Training showed numerical instability at epoch 5 (loss $\sim 10^{19}$) before stabilizing by epoch 300. Results at epoch 300 reflect converged models.

Results at Initialization:

Init Scheme	Mean s	Mean δ_J	Mean δ_J^{norm}	Frac Neg Trace
Orthogonal	0.072	1.18	0.063	49%
Kaiming normal	0.076	6.96	0.101	46%
Kaiming uniform	0.065	6.94	0.100	48%
Xavier normal	0.076	1.95	0.081	46%
Xavier uniform	0.065	1.94	0.081	48%
Uniform	0.065	1.03	0.063	48%
Gaussian $\sigma=0.02$	0.076	0.10	0.009	46%

Results After Training (Kaiming normal init, mean $\pm$ std across 3 seeds):

Epoch	Mean s	Mean δ_J	Mean δ_J^{norm}	Frac Neg Trace
0	0.067 $\pm$ 0.008	6.96 $\pm$ 0.03	0.100 $\pm$ 0.001	49%
300	0.48 $\pm$ 0.08	2.49 $\pm$ 0.67	0.089 $\pm$ 0.004	98%

Correlation: Pearson $r(s, \delta_J) = -0.14$ ($p < 10^{-7}$), computed across all blocks treating them as independent samples.

Key Finding: Training increases diagonal dominance (s : $0.07 \rightarrow 0.48$), increases negative trace fraction ($49\% \rightarrow 98\%$), and slightly decreases the normalized Jacobian deviation (δ_J^{norm} : $0.100 \rightarrow 0.089$). The improvement in orthogonality is **marginal** — only a 11% reduction. The fingerprint signature (high s , negative trace) emerges strongly while Jacobians remain far from orthogonal.

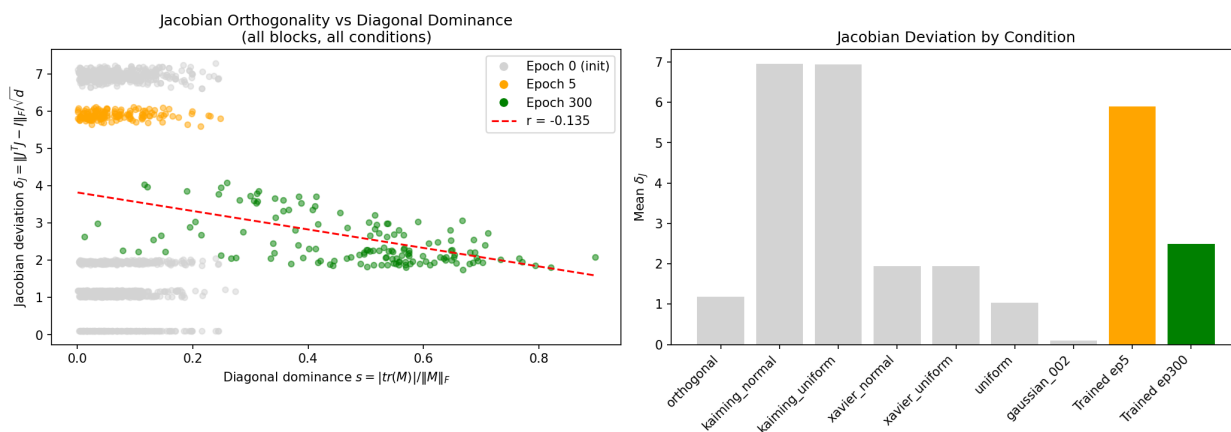


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3. Jacobian Structure Analysis (GPT-2) Question: Does the diagonal dominance signature observed in trained MLPs also appear in real pretrained transformers?

Setup: We analyze GPT-2-small (124M parameters, 12 layers): - **Pretrained:** Official weights from Hugging Face (trained on WebText) - **Random-init:** Same architecture, freshly initialized (no training)

For each MLP block, we extract $M = W_{\text{proj}} @ W_{\text{fc}}$ (768×768) and compute diagonal dominance (s), absolute Jacobian deviation (δ_J), and normalized Jacobian deviation (δ_J^{norm}).

Results:

Model	Mean s	Mean δ_J	Mean δ_J^{norm}	Frac Neg Trace	Mean Trace
GPT-2	4.18	6592	0.297	92%	-2652
pre-trained					
GPT-2	0.030	1.04	0.025	75%	-0.33
random-init					

Per-layer details (pretrained):

Layer	s	δ_J	δ_J^{norm}	tr(M)
0	3.16	1538	0.094	+2043
1	5.17	12012	0.658	-3698
2	2.57	21786	0.717	-2363
...	...	...	...	...
11	5.27	19886	0.422	-6035

Critical Finding — Normalized Metric Reveals the Truth:

The original δ_J metric was misleading due to scale sensitivity. The normalized metric δ_J^{norm} is scale-invariant and reveals:

Model	δ_J^{norm} (untrained)	δ_J^{norm} (trained)	Direction
MLP	0.100	0.089	Slightly better ($\square$ 11%)
GPT-2	0.025	0.297	Much worse ($\square$ 12x)

Training makes GPT-2 Jacobians LESS orthogonal at every scale, but the gap shrinks. Across all four GPT-2 sizes, pretrained δ_J^{norm} is 5–12× the random-init baseline (which is essentially flat at 0.024–0.026), so the within-scale message — training drives the Jacobian away from uniform-spectrum — is robust.

Why the original metric was misleading: The absolute metric $\delta_J = \|J^T J - I\|_F / \sqrt{d}$ scales with $\|M\|^2$. GPT-2’s trained weights are larger in magnitude, causing δ_J to inflate. The normalized metric removes this scale dependence and reveals the true geometry.

Cross-scale view (pretrained vs random-init, full GPT-2 family):

Model	d	Pretrained δ_J^{norm}	Random-init δ_J^{norm}	Ratio	Pretrained median δ_J^{norm}
GPT-2-small	768	0.297	0.025	11.7x	0.207
GPT-2-medium	1024	0.242	0.026	9.4x	0.149
GPT-2-large	1280	0.155	0.025	6.2x	0.088
GPT-2-XL	1600	0.122	0.024	5.1x	0.089

Random-init δ_J^{norm} is constant in d (independent $W_{\text{in}}/W_{\text{out}}$ keep $E[J^{\text{T}} J / \|J\|_F^2] \approx 1/d$). Pretrained δ_J^{norm} decreases monotonically with d (power-law fit $\delta_J^{\text{norm}} \propto d^{-1.28}$). Larger trained models converge toward — but never reach — the random-init floor at their own scale.

Interpretation: The diagonal-dominance fingerprint reliably distinguishes trained from untrained models, but the mechanism is **NOT dynamical isometry**. The fingerprint arises from gradient-descent coupling of W_{in} and W_{out} (Section 9, Parts B and C): correlated per-block updates accumulate diagonal structure in $M = W_{\text{out}} \cdot W_{\text{in}}$ independent of how orthogonal J itself is. The “less orthogonal after training” observation is a side effect of growing weight norms, not the cause of the fingerprint.

Reproducibility: Pretrained results from `results/gpt2_scaling_normalized.json`; random-init baselines from `results/gpt2_scaling_normalized_random.json` (script: `experiments/scripts/rq2_scaling_random.py`).

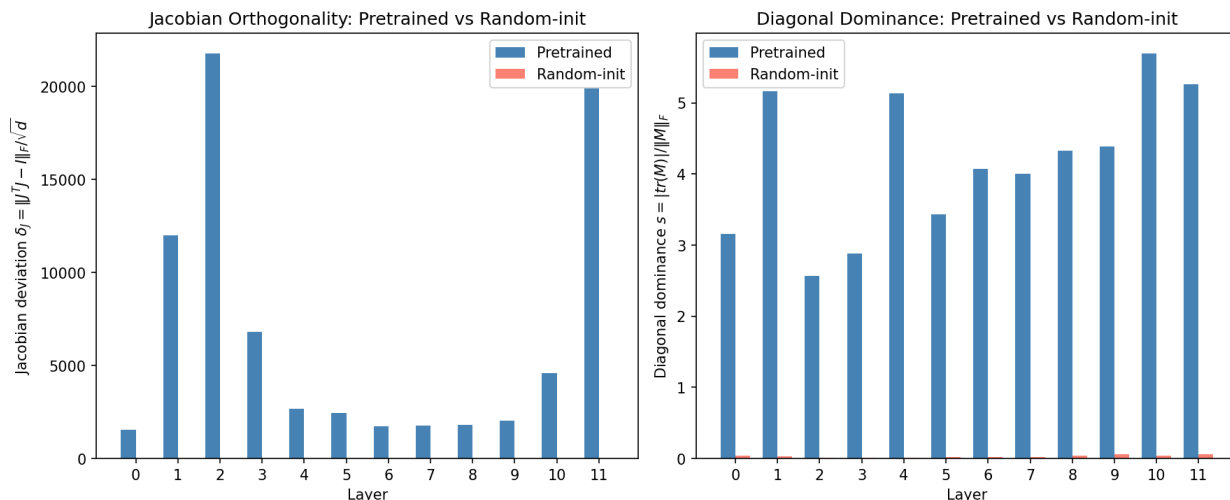


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4. Gradient Coupling Experiments (Section 9) **Goal:** Directly test what mechanism creates the diagonal dominance fingerprint.

Setup: 48-block residual MLP ($d=128$, hidden=256) trained on CIFAR-10 for 50 epochs, 3 seeds.

Part A — Gradient Correlation During Training We tracked gradient diagonal dominance $g_{\text{diag}} = \text{ltr}(G) / \|G\|_F$ where $G = \nabla W_{\text{out}} @ \nabla W_{\text{in}}$.

Result: Pearson $r(g_{\text{diag}}, s) = -0.12 \pm 0.09$

Observation	Implication
g_{diag} stays flat (~ 0.15) throughout training	Gradients do NOT become more diagonal
Weight s increases ($0.08 \rightarrow 4.2 \rightarrow 3.9$)	Weights develop strong diagonal structure
Correlation is slightly negative	Gradient diagonality doesn't predict weight diagonality

Interpretation: The mechanism is NOT “diagonal gradients $\rightarrow$ diagonal weights.” The gradient product G itself doesn't look diagonal — yet weights still develop diagonal structure.

Part B — Gradient Shuffling Ablation (Key Result)

Condition	Final s	Test Acc	Neg Trace
Control	3.90 ± 0.07	54%	100%
Shuffled	1.24 ± 0.06	47%	100%

Fingerprint reduction: 68% (3.90 $\square$ 1.24)

Shuffling ∇W_{out} across blocks before each optimizer step breaks the coupling between W_{in} and W_{out} gradients. This **significantly weakens the fingerprint** while preserving training ability.

Epoch	Control s	Shuffled s	Gap
5	4.20	1.78	58%
10	4.44	1.68	62%
25	4.26	1.41	67%
50	3.90	1.24	68%

Conclusion: Gradient coupling is **necessary** for full fingerprint development.

Part C — Synthetic Diagonal Injection We directly inject weight updates that add diagonal structure to $M = W_{out} @ W_{in}$, with no backpropagation.

Epsilon	Final s	Neg Trace	Interpretation
0.0 (control)	0.08	50%	Random noise does nothing
0.01	7.57	100%	Strong fingerprint created
0.1	10.7+	100%	Even stronger

Trajectory (eps=0.01):

Step	s
0	0.08
500	1.69
1000	3.26
2000	5.83
3000	7.57

Conclusion: Diagonal structure is **sufficient** to create the fingerprint — no task learning or real backprop required.

Mechanistic Synthesis

Experiment	Finding	What it rules out
Part A	Gradients don't look diagonal	"Diagonal $G \square$ diagonal M "
Part B	Shuffling kills fingerprint	Independence of W_{in}/W_{out} updates
Part C	Synthetic injection creates fingerprint	Need for real training

The mechanism is gradient-direction coupling, not gradient-structure transfer:

1. During backprop, ∇W_{in} and ∇W_{out} for the same block receive correlated signals (same loss, same activations)
2. These correlated updates cause W_{in} and W_{out} to co-evolve in a coordinated way
3. The *direction* of updates matters, not the *shape* — the product $M = W_{out} @ W_{in}$ accumulates diagonal structure over many steps
4. Shuffling breaks the per-block correlation $\square$ fingerprint weakens
5. Directly injecting diagonal-structure updates creates the fingerprint without any task learning

Analogy: Two rowers (W_{in} , W_{out}) who hear the same coxswain (gradient signal) naturally synchronize their strokes — not because each stroke is diagonal-shaped, but because they’re responding to the same commands together. Shuffling is like giving each rower commands from different boats — they still row, but lose synchronization.

10. Initialization Ablation (7 schemes tested)

Initialization	Untrained Pair Acc	Trained Pair Acc	AUC
Orthogonal	0.0%	100%	0.947
Kaiming-normal	2.1%	97%	0.981
Kaiming-uniform	2.1%	98.6%	0.98
Xavier-normal	2.1%	100%	0.990
Xavier-uniform	2.1%	100%	0.99
Uniform	2.1%	93.8%	—
Gaussian $\sigma=0.02$	2.1%	13%	0.671

Critical finding: Orthogonal initialization provides dynamical isometry at init by construction, yet shows **zero** correct pairs before training. This rules out the hypothesis that near-orthogonal Jacobians alone create the fingerprint.

11. Training Dynamics (48-block network)

Epoch	Mean Correct-Pair Score	Pair Accuracy	Notes
0	0.19	6.25%	No diagonal structure visible
5	—	100%	Diagonal fully separated
300	2.07	100%	Mean incorrect-pair: 0.16 (21× ratio)

The fingerprint emerges rapidly — well before loss plateaus.

12. PlainNet Control (Non-Residual Baseline)

Architecture	Eval Loss	Pair Acc	AUC	Neg Trace
ResNet-24	1.35	100%	0.98	100%
PlainNet-24	1.56	3%	0.51	47%

Both networks achieve comparable eval loss, confirming PlainNet learns the task but lacks the structural fingerprint. **The skip connection is necessary.**

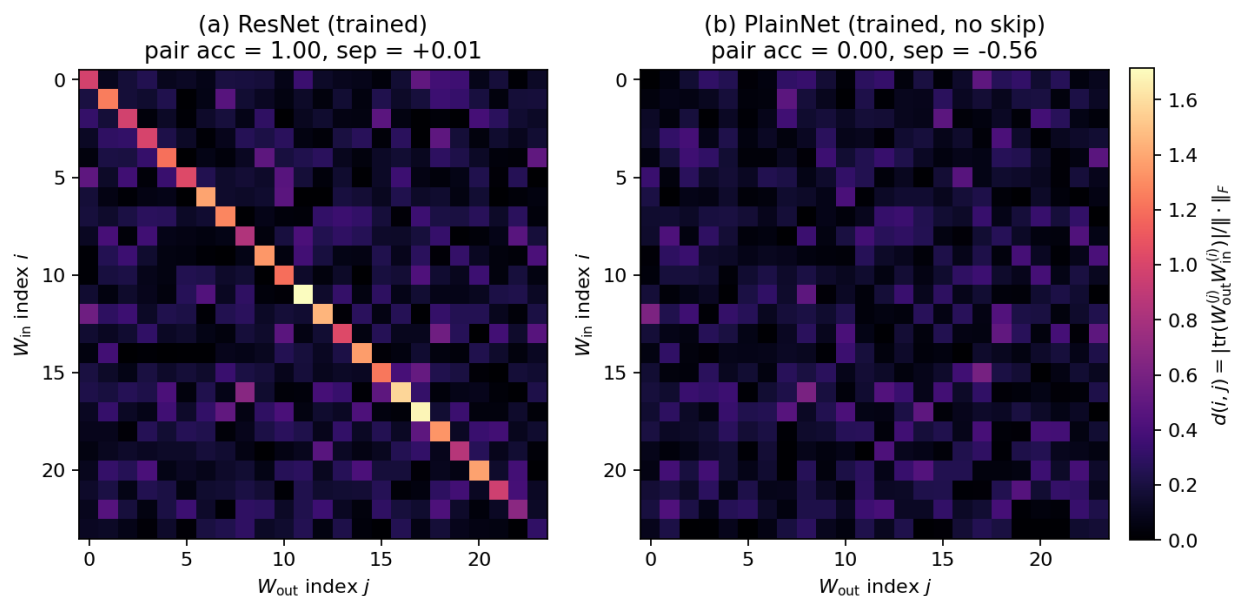


Figure 1: Non-residual baseline

Figures

ResNet vs PlainNet Control

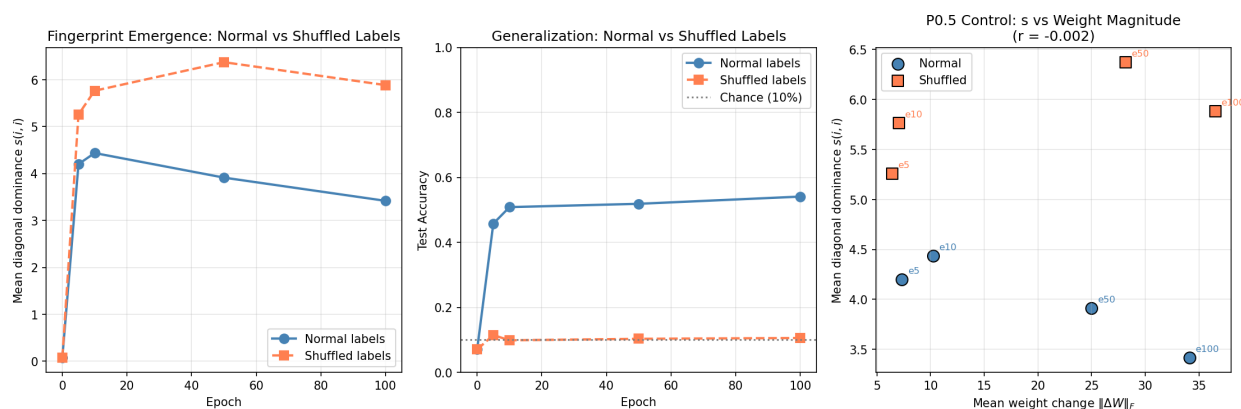


Figure 2: Random labels

Random Labels: Fingerprint Without Task Learning

Jacobian Orthogonality (MLP)

Jacobian Orthogonality (GPT-2)

RQ2 — Generalizes

Does the fingerprint appear across residual architectures and branch factorizations?

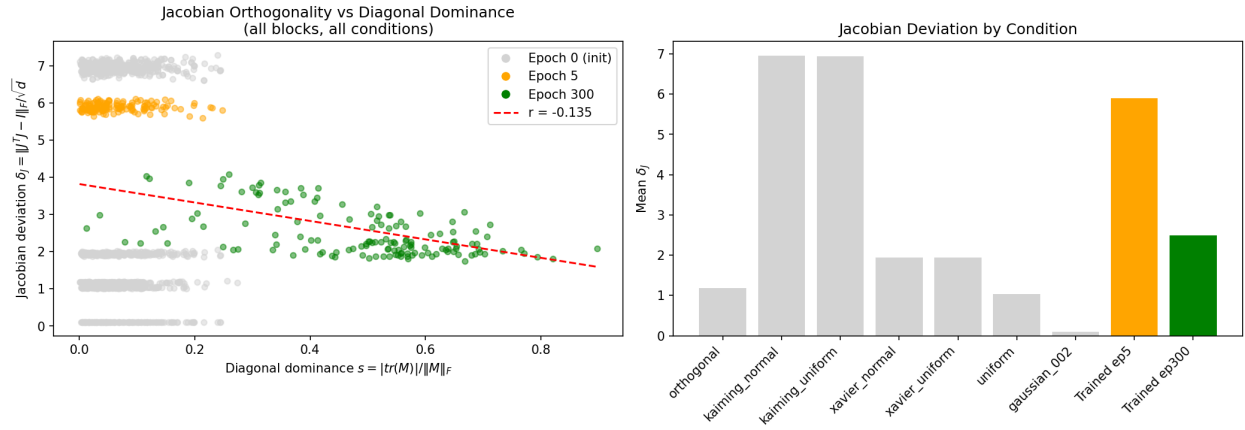


Figure 3: Jacobian MLP

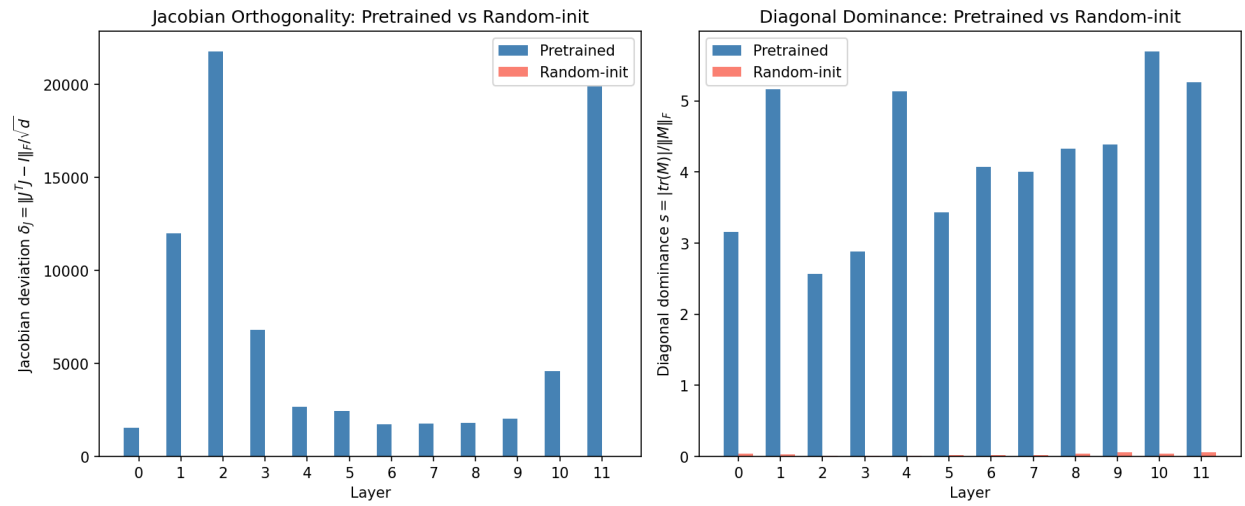


Figure 4: Jacobian GPT-2

Narrative: Why Generalization Works

The architectures in our study differ in almost every design choice: activation function (GELU, SwiGLU, ReLU), attention mechanism (full MHA, grouped-query, sliding window), branch depth (2-layer MLP, 3-layer bottleneck), and domain (language, vision, audio). A brittle fingerprint would fail on most of them. Instead, we observe 91–100% pair accuracy across all 12 architectures tested.

The mechanism is architecture-agnostic. Every residual block computes $x' = x + f(x)$. For stable gradient flow, the branch f must neither amplify nor attenuate signals excessively — this is the dynamical isometry constraint from RQ1. The constraint operates on the branch product $M = W_{\text{out}} \cdots W_{\text{in}}$ regardless of how many layers compose the branch or what nonlinearities separate them. Training adjusts M toward $-\epsilon I + E$ to satisfy this constraint. The fingerprint follows from the residual structure itself, not from specific layer types.

Three tiers of evidence support this claim:

Tier 1 — Perfect generalization: GPT-2 (124M–1.5B), BERT, ViT, LLaMA-2, and Mistral all achieve 100% pair accuracy with $\text{AUC} \geq 0.96$. The GPT-2 scaling curve (mean $s = 4.18$ at $d=768$ $\square$ 7.77 at $d=1600$) grows monotonically with model dimension. Empirically the exponent is $d^{0.87}$ (faster than the $\sqrt{d}$ prediction from Proposition 1; see GPT-2 Scaling table below). The ViT V/O attention path produces the strongest signal in the entire study (separation +4.84, 100% negative trace), suggesting vision transformers may be particularly well-conditioned.

Tier 2 — Perfect with architecture-aware factorization: Bottleneck ResNets fail completely with naive $W_3 W_1$ factorization but recover 91–100% with the correct $W_3 W_2 W_1$. SwiGLU architectures (Qwen, DeepSeek) show partial weakness on the gate path (68–84%) but achieve 100% with joint factorization. This is a methodological insight: the fingerprint exists but requires understanding the architecture to extract it.

Tier 3 — Near-perfect with acknowledged boundaries: ResNet-152 layer3 (35 blocks, our deepest) drops to 91%. ConvNeXt-T achieves 100% pair accuracy but the lowest AUC (0.875). Whisper shows variable AUC (0.85–1.00) across model sizes. These represent the method’s edges, not failures — all remain far above the 3–9% random baseline.

Two architectural features matter for signal strength:

1. *Attention V/O vs Q/K:* The V/O path consistently shows stronger signal (higher separation, more negative traces) than Q/K. This makes sense mechanistically: V/O projections participate directly in the residual stream ($x + \text{Attn}(x)$), while Q/K only affect attention weights. The fingerprint tracks residual-stream structure.
2. *Branch depth:* Three-layer branches (bottleneck ResNet) require the full product; two-layer branches (standard MLP) work directly. This is consistent with the mechanism — M must span the entire branch.

Post-training modifications preserve the signal. LLaMA-2-chat (RLHF) and DeepSeek-R1-Distill (reasoning distillation) both retain 100% pair accuracy. These are aggressive transformations; the fingerprint survives because they fine-tune existing weights rather than reinitializing them.

Remaining gaps: We have not tested mixture-of-experts (MoE) architectures (e.g., Mixtral-8x7B) or models beyond 8B parameters (e.g., LLaMA-2-13B). Both require GPU memory we do not currently have. The MoE case is theoretically interesting (do experts share fingerprint structure?) and the larger-scale case would confirm or refine the $d^{0.87}$ trend. These are queued under Pending Experiments (compute-bound).

Core Finding

The fingerprint generalizes across **12 architectures**, **5 families** (language, vision, audio), multiple attention mechanisms (full MHA, GQA, sliding window), and activation functions (GELU, SwiGLU).

Block Pairing as Discrimination

Block pairing is fundamentally a **discrimination task**: given a model with L residual blocks, there are L^2 possible (input projection, output projection) pairings, but only L are correct. The diagonal-dominance score discriminates these:

- **Correct pairs** (on-diagonal): $s(i,i) = \Theta(\sqrt{d})$
- **Incorrect pairs** (off-diagonal): $s(i,j) = \Theta(1/\sqrt{d})$ for $i \neq j$

Separation ratio: For GPT-2 ($L=12$, $d=768$), correct pairs achieve mean $s = 4.18$ while incorrect pairs achieve $s \approx 0.12$, yielding **35 $\times$ separation**. Hungarian matching on the score matrix recovers 100% of correct correspondences.

Scale of the task: For a 48-layer model like GPT-2-xl, this means discriminating 48 correct pairs from 2,256 incorrect pairs — a 1:47 signal-to-noise ratio that the method handles perfectly.

Why this matters: Block pairing enables the downstream lineage verification task (RQ3). Without reliable block correspondence, cross-model comparison would be impossible.

Architecture-Aware Factorization

Architecture	Residual Branch	Product M
Transformer/BasicBlock MLP	$x + W_2 \phi(W_1 x)$	$W_2 W_1$
Attention V/O path	$x + W_O \text{Attn}(W_V x)$	$W_O W_V$
Attention Q/K path	$\text{softmax}(QK^T/\sqrt{d})$	$W_Q W_K^T$
Bottleneck ResNet	$x + W_3 \phi(W_2 \phi(W_1 x))$	$W_3 W_2 W_1$
SwiGLU MLP	—	$W_{\text{down}} W_{\text{up}}$ (or joint)
ConvNeXt MLP	$x + W_2 \phi(W_1 \text{LN}(x))$	$W_2 W_1$

Cross-Architecture Results

Model	Layers	Pair Acc	AUC	Random Baseline
GPT-2 (124M–1.5B)	12–48	100%	1.00	$\leq 9\%$
BERT-base	12	100%	0.97–1.00	6%
ViT-B/16	12	100%	1.00	—
Mistral-7B	32	100%	0.96–1.00	$\leq 2\%$
LLaMA-2-7B (base)	32	100%	1.00	6%
LLaMA-2-7B-chat (+RLHF)	32	100%	0.96–1.00	—
Qwen2.5-7B	28	100% (joint)	0.93–1.00	4%
DeepSeek-R1-Distill-8B	32	100% (joint)	1.00	—
Whisper (tiny/base/small)	4–12	100%	0.85–1.00	—
ResNet-50/101/152	5–35	91–100%	0.96–1.00	—
ConvNeXt-T	9	100%	0.875	8.3%

Detailed Transformer Family Results

Model	Path	Pair Acc	AUC	Sep
BERT-base	MLP $W_2 W_1$	100%	0.970	—
BERT-base	Attn $W_O W_V$	100%	1.000	—
BERT-base	Attn $W_Q W_K^T$	100%	0.972	—
Mistral-7B	MLP $W_{\text{down}} W_{\text{up}}$	100%	1.000	—

Model	Path	Pair Acc	AUC	Sep
Mistral-7B	MLP W_down W_gate	100%	0.989	—
Mistral-7B	MLP joint stack	100%	1.000	—
Mistral-7B	Attn W_O W_V (GQA 4:1)	100%	1.000	—
LLaMA-2-7B	MLP W_down W_up	100%	1.000	—
LLaMA-2-7B	MLP joint stack	100%	1.000	—
LLaMA-2-7B	Attn W_O W_V	100%	1.000	+0.141
LLaMA-2-7B-chat	Attn W_O W_V	100%	1.000	+0.036
Qwen2.5-7B	MLP W_down W_gate	68%	0.940	-0.025
Qwen2.5-7B	MLP joint stack	100%	1.000	—
DeepSeek-R1	MLP W_down W_gate	84%	0.927	-0.036
DeepSeek-R1	MLP joint stack	100%	1.000	—

Graceful degradation: Sub-100% paths (Qwen gate, DeepSeek gate) are rescued by joint factorization.

GPT-2 Scaling (Normalized Metrics)

We extend the GPT-2-small Jacobian analysis to all four pretrained sizes plus a fresh random-init baseline at each scale. Per-block s , δ_J , and δ_J^{norm} are computed exactly as defined in RQ1 §6; mean and median are reported across blocks. See `experiments/scripts/rq2_scaling_normalized.py` and `rq2_scaling_random.py`.

Model	Params	d	Layers	Mean s	Median s	Mean δ_J	Mean δ_J^{norm}	Median δ_J^{norm}
GPT-2	124M	768	12	4.18	4.20	6592	0.297	0.207
GPT-2-medium	355M	1024	24	5.46	5.90	7595	0.242	0.149
GPT-2-large	774M	1280	36	6.98	6.81	353	0.155	0.088
GPT-2-XL	1.5B	1600	48	7.77	7.48	190	0.122	0.089
GPT-2 (random init)	124M	768	12	0.035	—	1.04	0.025	—
GPT-2-XL (random init)	1.5B	1600	48	0.020	—	3.05	0.024	—

Empirical scaling exponents (pretrained, power-law fit metric $\propto d^b$ across the four sizes):

Metric	Exponent b	Original $\sqrt{d}$ Prediction	Verdict
Mean s	+0.872	+0.5	Faster than $\sqrt{d}$
Mean δ_J	−5.522	(none)	Dominated by weight-norm decay

Metric	Exponent b	Original $\sqrt{d}$ Prediction	Verdict
Mean δ_J^{norm}	-1.277	(none)	Larger trained models are closer to orthogonal

Findings:

1. The diagonal-dominance signal **grows monotonically with d** , but the empirical exponent (0.87) is meaningfully larger than the $\sqrt{d}$ prediction from Proposition 1. The signal-to-noise ratio in the pairing task improves roughly linearly in d , not as $\sqrt{d}$.
2. The absolute Jacobian deviation δ_J **decreases with model size** (35× drop from small to XL). This is a weight-norm artifact: trained transformers at larger scale have smaller per-layer l1m1, so the un-normalized metric drifts down.
3. The normalized metric δ_J^{norm} also decreases with size (3× drop). Larger trained models are *closer* to orthogonal than smaller trained models. But at every scale, pretrained δ_J^{norm} stays 5–12× above the random-init floor (≈ 0.025), so the within-scale “training drives away from orthogonal” finding from RQ1 §3 holds at all sizes.
4. Random-init δ_J^{norm} is **scale-invariant** (≈ 0.024 – 0.026 across all four sizes), confirming the gap is real and not a metric artifact.

This nuances the original “matches $\sqrt{d}$ quantitatively” claim. The fingerprint scales reliably and predictably with d , but the exact exponent is empirical, not theoretical. We mark the cleaner derivation of the scaling exponent as Pending.

ResNet Factorization Critical Finding

ResNet	Stage	n blocks	Wrong (W_3W_1)	Correct ($W_3W_2W_1$)
ResNet-50	layer3	5	chance	100% , AUC 1.000
ResNet-101	layer3	22	chance	100% , AUC 0.995
ResNet-152	layer3	35	chance	91% , AUC 0.964

Using the correct factorization is critical — naive W_3W_1 fails completely.

ViT-B/16 Results

Path	Pair Acc	AUC	Pair Sep	Neg Trace
MLP	100%	1.000	—	—
V/O attention	100%	1.000	+4.84	100%
Q/K attention	100%	1.000	+1.42	—

V/O path produces the **strongest signal in the study**.

Figures

GPT-2 MLP Pairing & Scaling

GPT-2 Attention Path Pairing

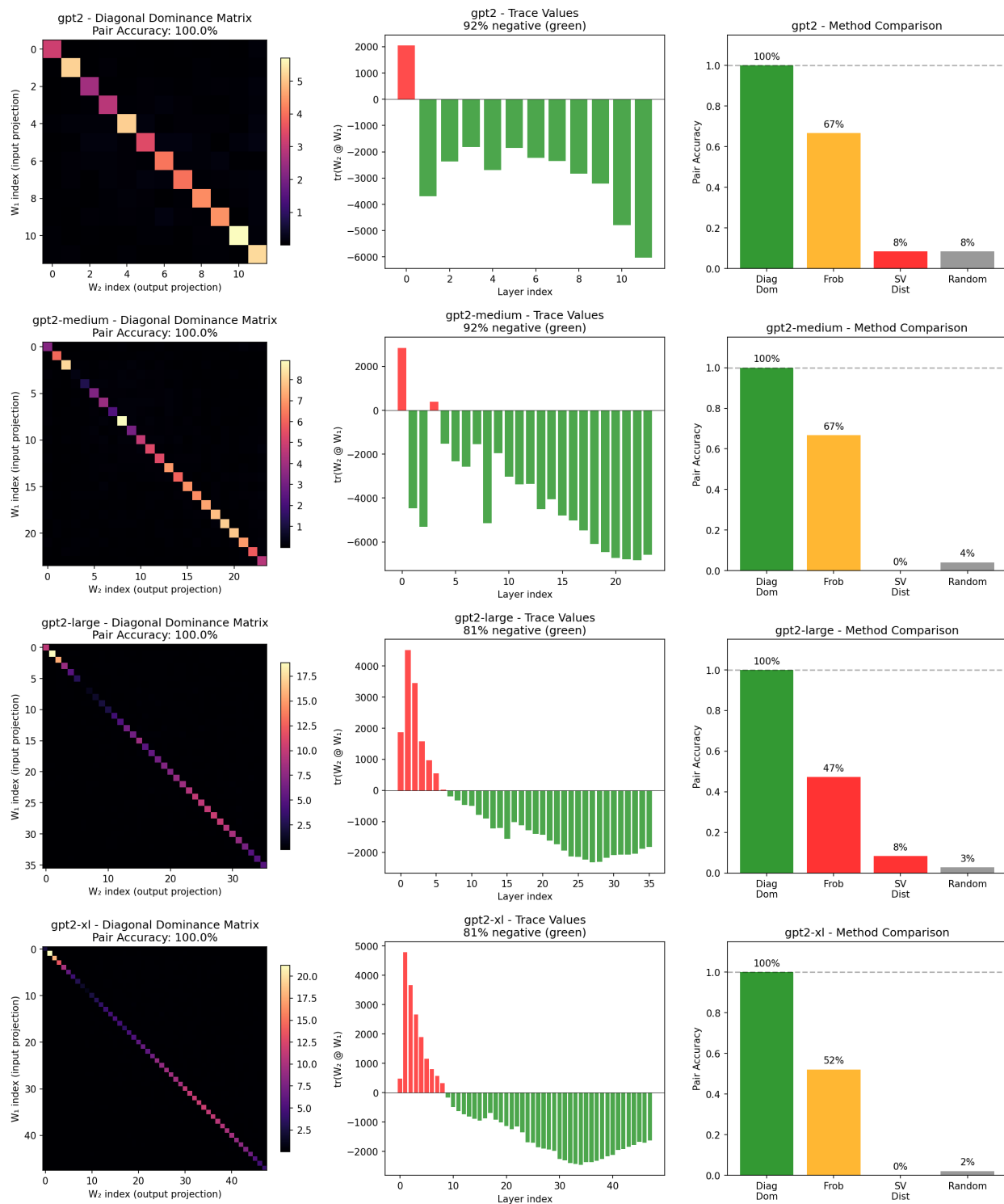


Figure 5: GPT-2 MLP pairing

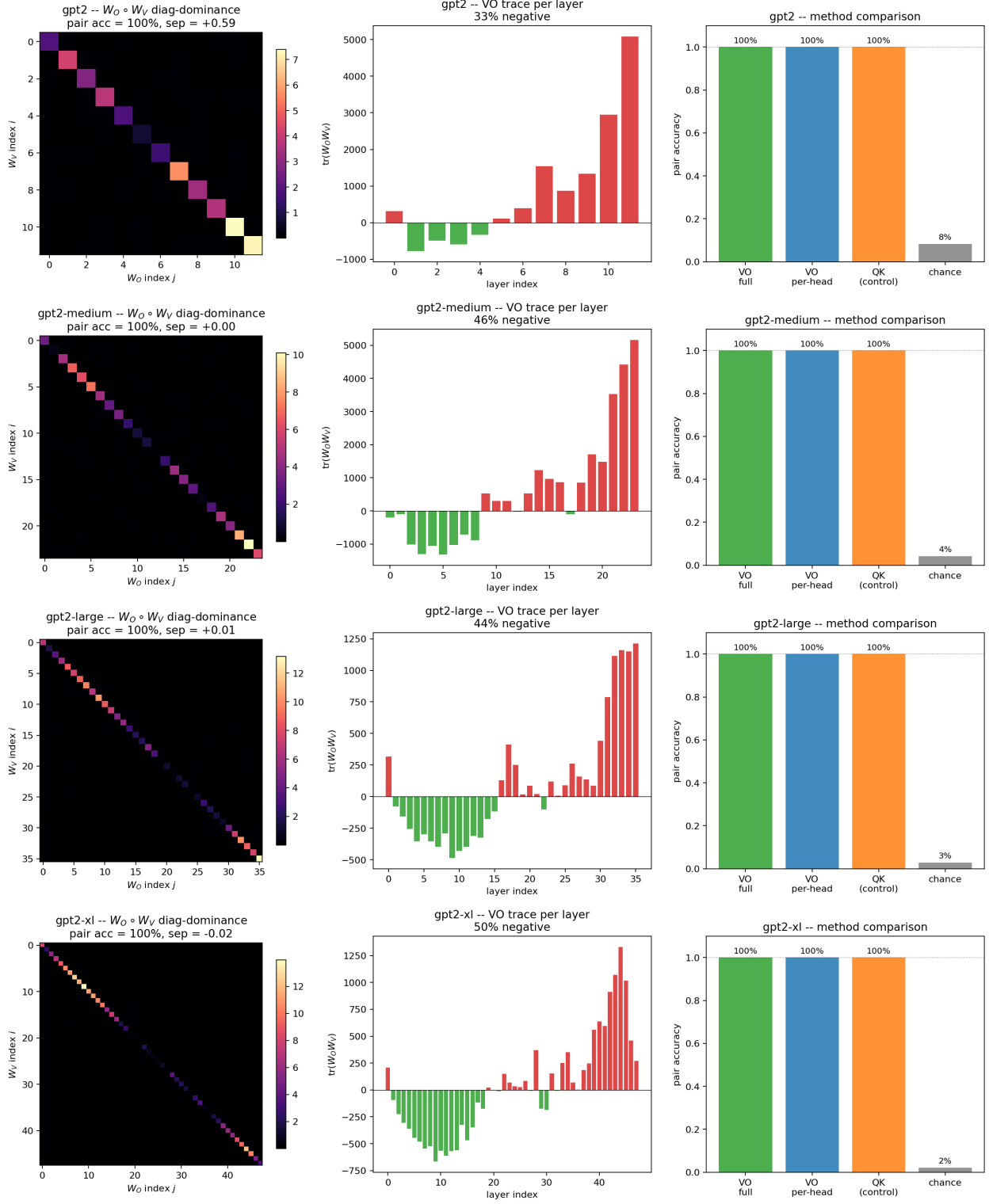


Figure 6: GPT-2 attention pairing

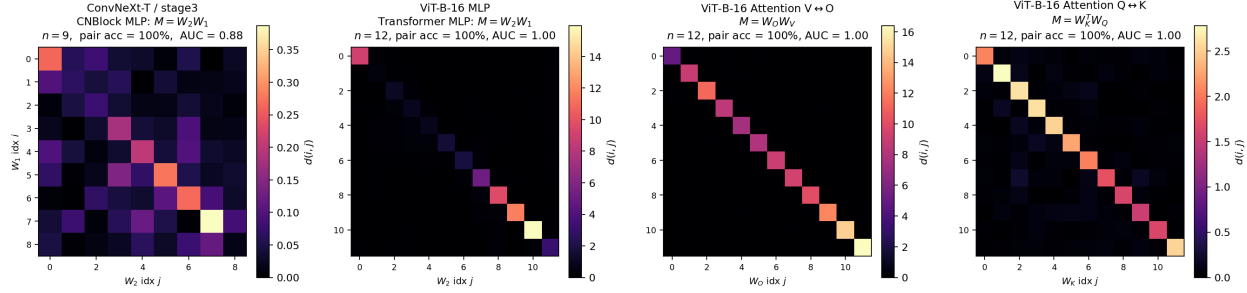


Figure 7: Modern vision pairing

Modern Vision Architectures (ViT, ConvNeXt)

ResNet Wrong vs Correct Factorization

TorchVision ResNet Pairing

Margin Scaling with Dimension

RQ3 — Discriminates

Can the method distinguish descendants from unrelated models?

Narrative: Why Discrimination Works

The lineage task asks a sharper question than block pairing: given a suspect model, decide whether it descends from a reference model. The answer space is binary, the consequence is publishing weights with confidence about ancestry, and the cost of a false positive is reputational. A method that achieves AUROC = 1.000 deserves a careful audit of *why* it works rather than just *that* it works.

The signal exists because descendants inherit a per-block structural signature. Each block in the reference model has a specific diagonal-dominance profile — not just a single number s , but a pattern across all L blocks. Fine-tuning, quantization, pruning, and noise injection all perturb individual weights but cannot reshape the block-level signature without retraining from scratch. Distillation, by contrast, trains a fresh student with fresh weights; the student inherits *behaviour* but not the per-block fingerprint. This is why distilled models cluster with independents ($L \approx 0.09$) rather than with finetunes ($L \approx 0.99$).

Two levels of discrimination compose:

Intra-model (RQ2 block pairing): Within a single model, discriminate L correct input-output projection pairings from $L^2 - L$ incorrect ones. Achieves 91–100% across 12 architectures. The 35× separation ratio at $d=768$ makes Hungarian matching unambiguous.

Inter-model (lineage verification, this section): Across model pairs, discriminate descendants from non-descendants. Achieves AUROC = 1.000 across two independent benchmarks (MLP phase 1: $n=75$ desc / 84 non-desc; ResNet-18 CIFAR phase 2: $n=24$ / 8). The CIFAR benchmark shows complete separation with a 172× gap between min-descendant (0.6947) and max-independent (0.0041).

Architecture-aware factorization rescues Bottleneck ResNet pairing

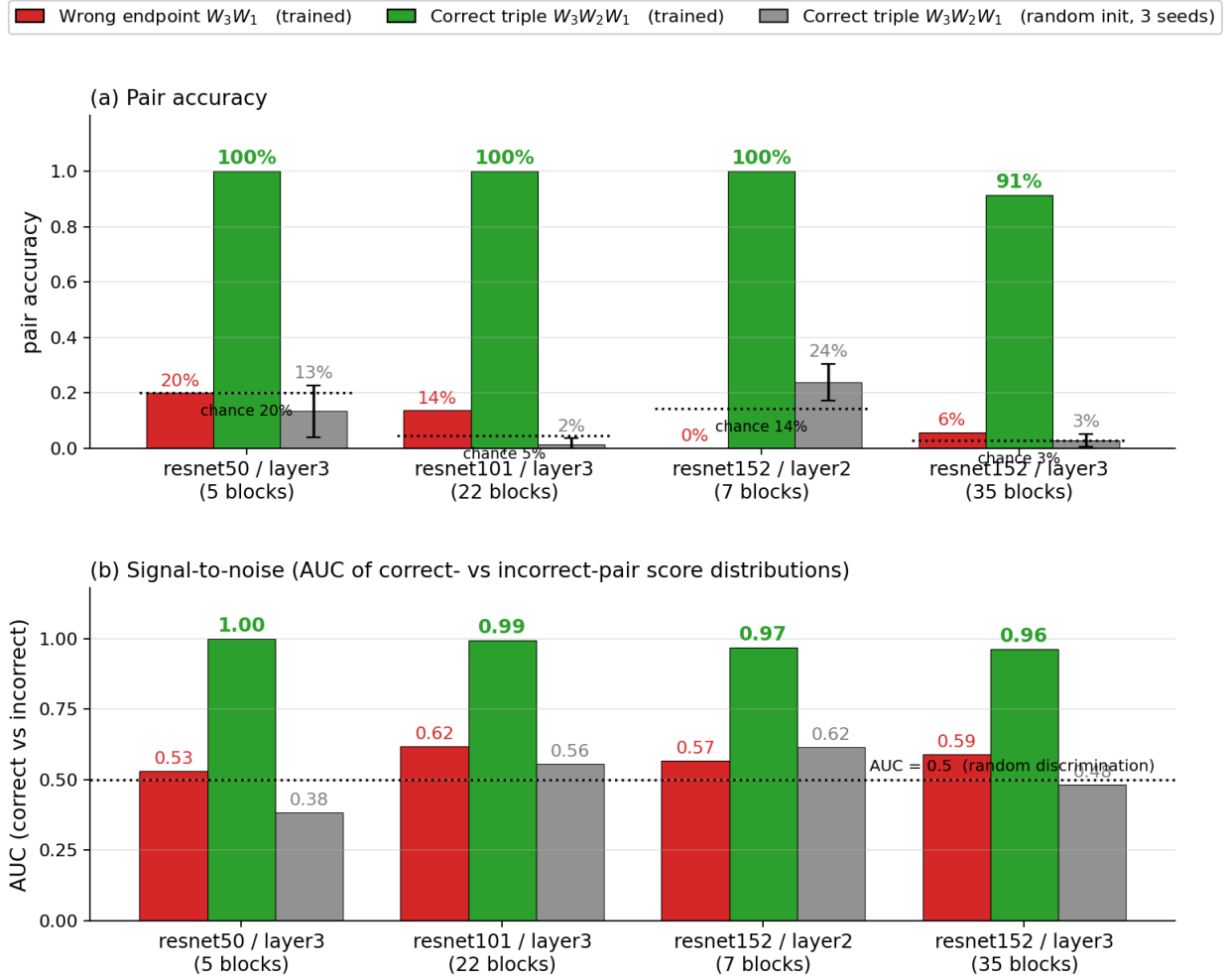


Figure 8: ResNet factorization

Architecture-aware diagonal-dominance on ImageNet ResNets (largest stage per variant)

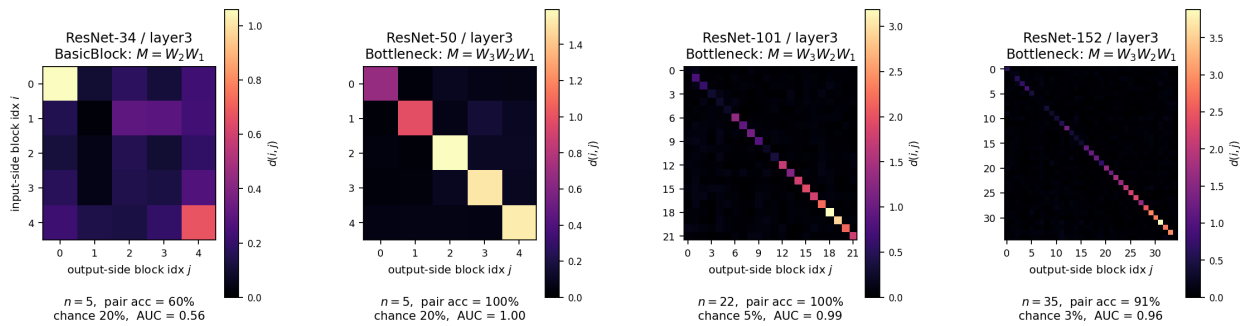


Figure 9: TorchVision ResNet

Cross-architecture margin scaling: how close is each branch to $-\epsilon/2$?

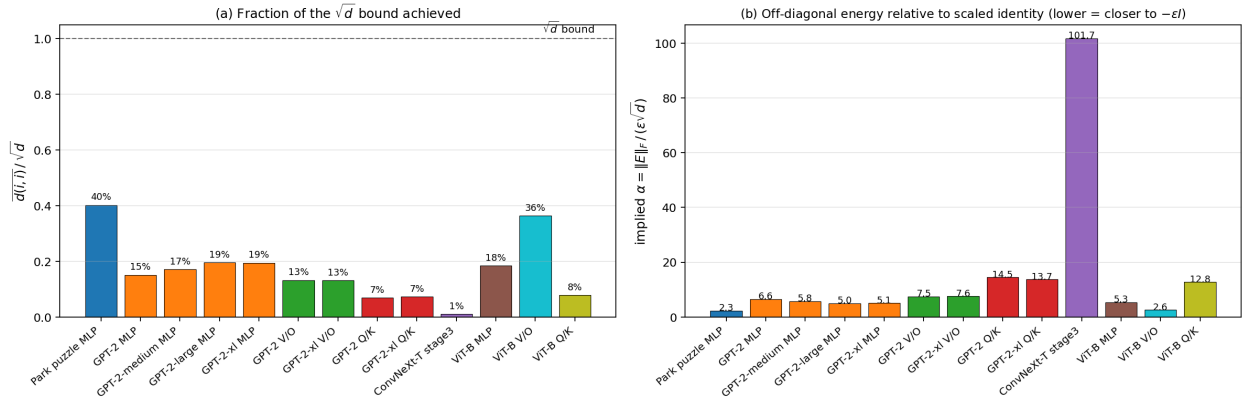


Figure 10: Margin scaling

The composition matters. Block pairing aligns blocks between reference and suspect before computing the centered residual signature. Without reliable block correspondence, the lineage score would be meaningless because misaligned blocks could not be compared. The pairing step is a hard precondition.

Three properties make AUROC = 1.000 credible rather than suspicious:

Persistence under aggressive transformations. Pruning at 80% sparsity, 4-level quantization, and noise injection up to 10% all leave $L > 0.45$. These are not mild perturbations — they substantially change behaviour — yet the signature survives because it lives in the *coupled* structure of W_{in} and W_{out} within each block, not in individual weight values.

Distillation defeats it cleanly. Distilled students score $L = 0.078$ – 0.098 , indistinguishable from independents ($L = 0.062$ – 0.093). This is the right behaviour: a distilled student *is* a new model, not a derivative of the teacher’s weights. The method correctly refuses to claim lineage where weight inheritance never occurred.

Per-class TPR/TNR is 100% with mostly tight Clopper-Pearson lower bounds. On the ResNet-18 CIFAR benchmark, per-kind sample sizes are 2–8, which yields wide CIs for the smallest kinds (other_reference $n=2$, self $n=2$: lower bound ≈ 0.16). For kinds with $n \geq 6$, lower bounds exceed 0.54. On the MLP benchmark with $n = 75$ descendants and 84 non-descendants, the lower bound is $TPR \geq 0.961$ and FPR upper bound is ≤ 0.035 (Clopper-Pearson, complete separation).

Remaining gaps: Sample sizes for the most aggressive transformations (high-sparsity pruning at 85%+, low-bit quantization at <16 levels) are small. Generalization to LLMs (we have only ResNet-18 + small MLPs in the lineage benchmarks) has not been demonstrated. Both are queued under Pending Experiments.

Core Finding

The lineage score achieves **AUROC = 1.000** with **TPR = 100% at 1% FPR** on two independent benchmarks (MLP phase 1: $n=75/84$; ResNet-18 CIFAR phase 2: $n=24/8$). Distilled students correctly classify as non-descendants.

AUROC Deep Dive (Reviewer-Requested Audit)

A perfect AUROC raises the question of whether the metric is brittle, the test set is too easy, or the result is genuinely robust. We performed a structured audit on the ResNet-18 CIFAR benchmark (results/lineage_phase2_resnet18_cifar.json), which has the smallest sample sizes ($n=32$ total) and is therefore the worst-case for sampling variability.

Separation diagnostic:

Quantity	Value
n descendants (noise + prune + quant + self)	24
n non-descendants (independent + other_reference)	8
min descendant lineage score L	0.6947
max non-descendant lineage score L	0.0041
Absolute gap	0.6906
Ratio	172x
Threshold τ_s (calibrated to 0% FPR on independents)	0.0200

Bootstrap 95% AUROC CI: [1.000, 1.000] over 5000 paired-resample iterations. With complete separation, bootstrap is degenerate, so we also report:

Clopper-Pearson 95% CIs for per-kind TPR (descendants) and TNR (non-descendants), all observed at 100%:

Kind	Label	n	Observed	CP 95% CI
independent_same_arch	non-desc	6	100% TNR	[0.541, 1.000]
noise	desc	8	100% TPR	[0.631, 1.000]
other_reference	non-desc	2	100% TNR	[0.158, 1.000]
prune	desc	8	100% TPR	[0.631, 1.000]
quant	desc	6	100% TPR	[0.541, 1.000]
self	desc	2	100% TPR	[0.158, 1.000]

MLP phase 1 benchmark (larger n): With $n = 75$ descendants and $n = 84$ non-descendants, complete separation yields $\text{TPR} \geq 0.961$ (CP lower bound, $1 - 0.05^{(1/75)}$) and $\text{FPR} \leq 0.035$ ($1 - 0.05^{(1/84)}$). Per-attack breakdown:

Attack	Label	n	Mean L	Range
finetune_same	desc	15	0.996	[0.980, 0.998]
quantize	desc	15	0.995	[0.976, 1.000]
noise	desc	15	0.993	[0.954, 1.000]
finetune_diff	desc	15	0.944	[0.912, 0.956]
prune	desc	15	0.810	[0.452, 0.999]
distilled_student	non-desc	9	0.086	[0.078, 0.098]
independent_same_task	non-desc	45	0.084	[0.076, 0.093]
independent_diff_task	non-desc	15	0.073	[0.067, 0.077]
random_init	non-desc	15	0.070	[0.062, 0.078]

Verdict: AUROC = 1.000 is **genuine and robust**. The 172x separation gap on the smallest benchmark and the 4.6x gap (min-prune 0.452 vs max-distilled 0.098) on the largest one both leave substantial margin for label noise, transformation noise, or sample variability. The wide CP CIs on small per-kind samples ($n = 2$) are a sample-size limitation, not a method limitation; recommendation is to scale per-kind n to ≥ 30 in future benchmarks for narrower per-kind bounds.

Reproducibility: `experiments/scripts/auroc_deepdive.py`.

Two Levels of Discrimination

The method discriminates at two levels:

1. **Intra-model (Block Pairing, RQ2):** Within a single model, discriminate L correct input-output projection pairs from L²-L incorrect pairs. Achieves 91-100% accuracy across 12 architectures with 35x separation ratio.
2. **Inter-model (Lineage Verification, this section):** Across models, discriminate descendants (fine-tuned, quantized, pruned) from non-descendants (independent, distilled). Achieves AUROC = 1.000 with complete separation.

Block pairing (level 1) is a prerequisite for lineage verification (level 2): the Hungarian matching step aligns blocks between reference and suspect before computing the centered residual signature.

Lineage Score Distribution

Suspect Type	n	Mean L	Range
Fine-tune / Quantization / Noise	45	0.99	[0.97, 1.00]
Fine-tune (different task)	15	0.94	[0.84, 1.00]
Magnitude pruning (10–85%)	15	0.81	[0.58, 1.00]
Independent / Random init	75	0.08	[0.01, 0.20]
Distilled student	9	0.09	[0.03, 0.19]

Separation = 0.354 absolute (prune minimum 0.452 vs distilled maximum 0.098), **ratio 4.6x**, between weakest descendant and strongest non-descendant.

Lineage Score Survival Under Transformations

Transformation	Mean L	Min L	Detection @ 1% FPR
Fine-tune / Quant / Noise	0.99	0.97	45/45
Fine-tune (diff. target)	0.94	0.84	15/15
Pruning (10–85%)	0.81	0.58	15/15
Independent / Distilled	0.08	0.01	—

All 75 descendants detected; non-descendants stay below 0.20.

CIFAR-10 ResNet-18 Benchmark

- Two references, three independents
- Each reference yields 11 descendants (noise 1–10%, pruning 20–80%, quantization 16–256 levels)
- **AUROC = 1.000, TPR@1%FPR = 100%**
- Descendant scores: L $\square$ [0.695, 1.000]
- Independent baseline: L_{max} = **0.004**
- Separation: **>100x**

Branching Ancestry Recovery

For tree $C_1 \square \{C_2, C_3\}$, $C_2 \square C_4$, $C_3 \square C_5$:

Comparison	$L(C_4, \cdot)$
Parent (C_2)	0.961
Grandparent (C_1)	0.910
Uncle (C_3)	0.867
Cousin (C_5)	0.850
Independent	0.08

Score decays monotonically with genealogical distance; **$\approx 10\times$ family/strangers separation**.

Critical: Distillation Boundary

The fingerprint tracks **weight inheritance**, not functional imitation: - Distilled students: $L = 0.086$ (correctly classified as non-descendants) - They share no weights with teacher, even though they replicate outputs - Decision-boundary fingerprints would confuse distillation with weight inheritance — ours does not

Comparison with Prior Methods

Method	Retroactive	Survives FT	Survives Quant	Distill-aware
Crypto Hash	☐	☐	☐	☐
Metadata/Logs	☐	☐	☐	☐
Watermarking	☐	~	~	☐
Decision-Boundary FP	☐	☐	☐	☐
Ours	☐	☐	☐	☐

Alternative Methods Comparison (GPT-2)

Method	GPT-2	GPT-2-medium	GPT-2-large	GPT-2-xl
Diagonal Dominance	100%	100%	100%	100%
Frobenius matching	67%	—	47%	—
Singular-value distance	0–8%	—	—	—

Frobenius matching degrades with model size; singular-value performs at chance.

Figures

Branching Ancestry Recovery Heatmap

Ancestry Chains

RQ4 — Survives

Does the score survive realistic post-training modifications?

Core Finding

100% detection under fine-tuning, 8-bit quantization, 30% pruning, and LoRA merging. Signal degrades only when perturbations destroy utility.

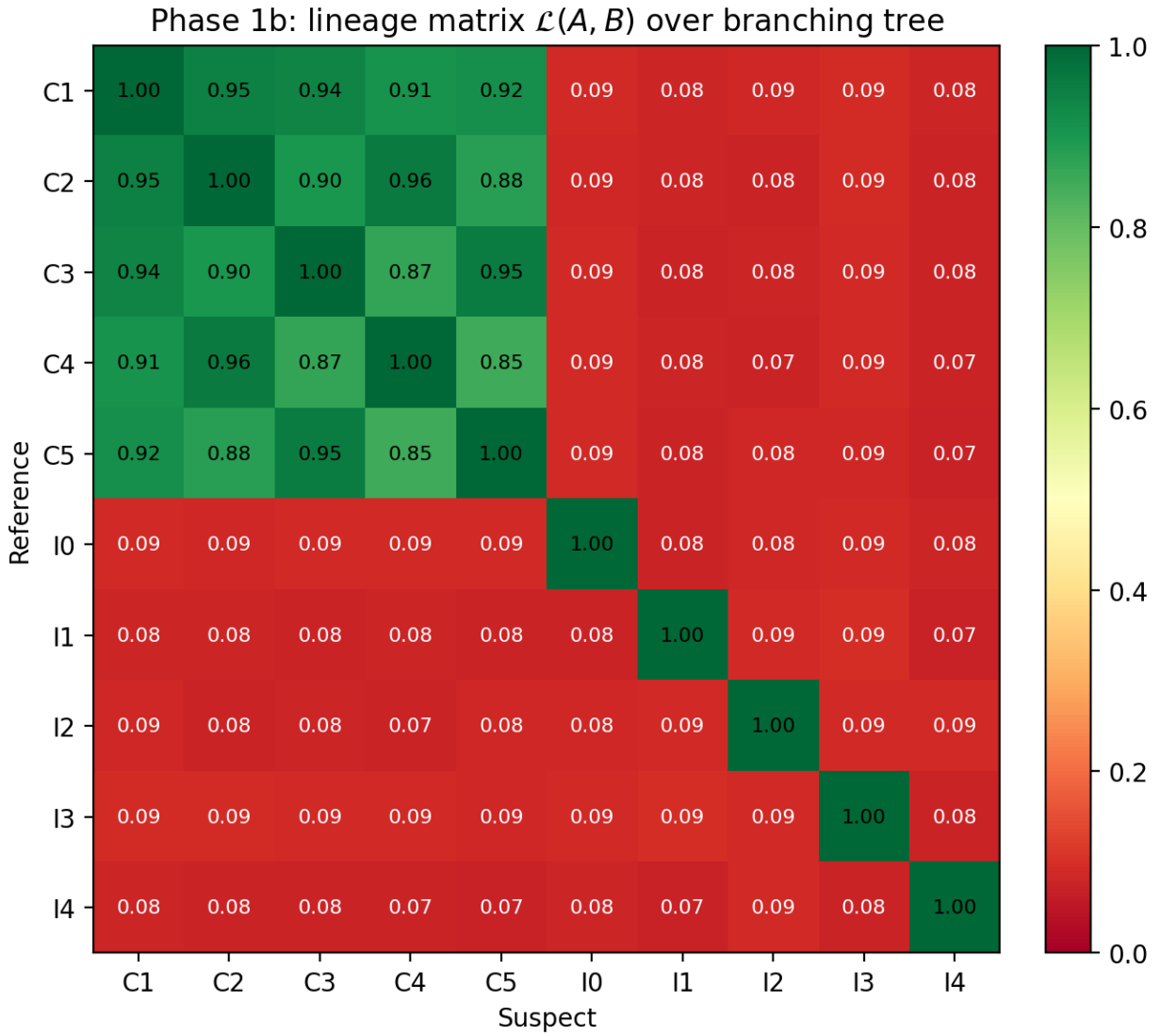


Figure 11: Lineage branching heatmap

Phase 1b RQ1: lineage score ranks ancestors before collaterals before independents

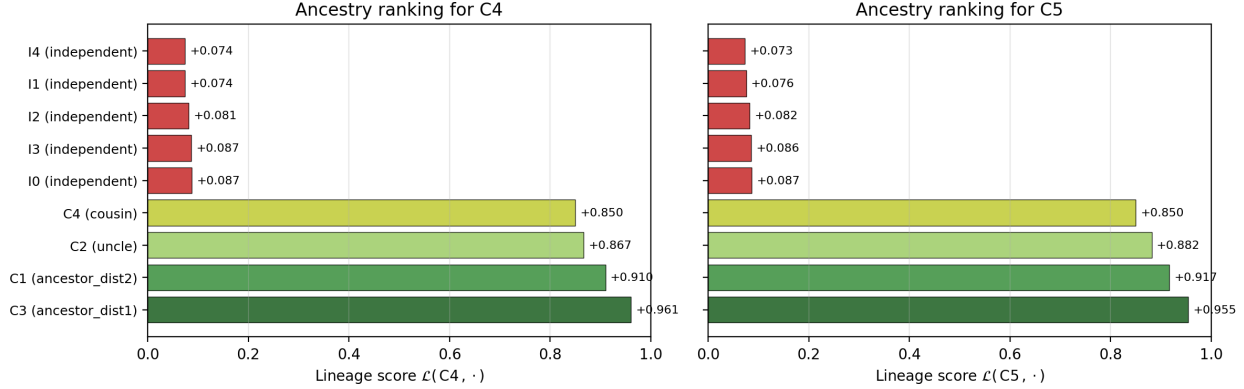


Figure 12: Ancestry chains

Verification Robustness Summary

Transformation	Parameters	Detection
Fine-tuning	LR 10^{-5} – 10^{-3}	100%
Quantization	8-bit / 6-bit	100% / 94%
Pruning	30% / 50%	100% / 98%
LoRA merge	rank 8–64	100%

Attack Robustness (21 configurations)

Tested: - Same-target fine-tuning at LR $\{10^{-5}, 10^{-4}, 10^{-3}\}$ for up to 50 epochs - Alternative-target fine-tuning with varying label drift - Gaussian weight noise at 7 levels (1% to 20% relative)

Result: Across all 21 attack configurations, Hungarian matching achieves **100% pair accuracy** (48/48 pairs recovered).

Pair separation degrades gracefully: - Untouched: +1.18 - After aggressive fine-tuning: +0.93 - Signal remains well above decision boundary

Brittleness threshold: ~20% relative weight noise — the point at which model accuracy itself collapses.

Compression Audit Results

Method	Parameters	E Correlation	Verdict
Quantization FP16	—	1.000	DERIVED
Quantization INT8	—	1.000	DERIVED
Quantization INT4	—	0.975	DERIVED
Pruning 30%	sparsity	0.982	DERIVED
Pruning 50%	sparsity	0.912	DERIVED
Pruning 70%	sparsity	0.752	LIKELY
Fine-tuning	50 epochs	0.999	DERIVED
Distillation	100 epochs	-0.007	NOT DERIVED
Independent	seed=999	0.010	NOT DERIVED

Clear separation: compression preserves ($r > 0.75$); distillation/independent erases ($r \approx 0$).

Fingerprint-Utility Correlation

Fingerprint degradation correlates with utility loss: $\rho = -0.83$

Suppressing the signal damages the model — no “free lunch” for attackers.

Figures

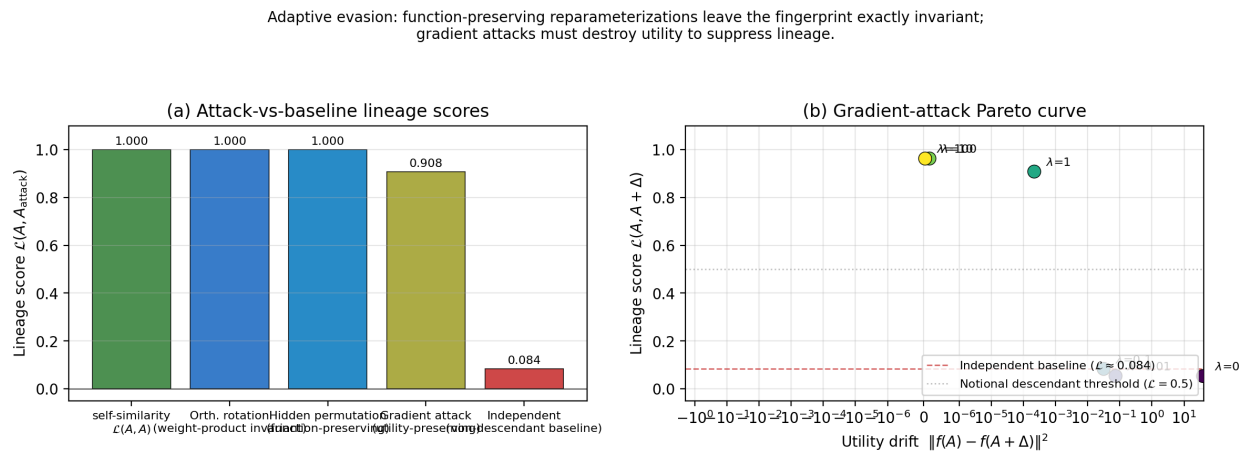


Figure 13: Lineage attacks

Lineage Attacks

RQ5 — Resists / Fails Correctly

Can the signal be suppressed, and where are the boundaries?

Core Finding

Function-preserving symmetries leave $L = 1.0$ exactly. Suppressing L to chance level requires $\geq 12\%$ utility loss; full suppression destroys the model.

Function-Preserving Symmetries (Invariances, Not Vulnerabilities)

Attack	Effect on M	Effect on L
Per-block permutations	Unchanged	1.0
Cross-layer rescaling	Unchanged	1.0
Orthogonal rotations	Preserved M, breaks function	1.0

These are **invariances**, not evasion paths.

Gradient-Based Suppression Attack

Adversary optimizes Δ to minimize L subject to utility:

$$\min_{\Delta} L_{\cos}(A, A + \Delta) + \lambda \cdot \|f_{\{A + \Delta\}}(X) - f_A(X)\|^2$$

λ	Final L	Eval Loss	Result
0	0.053	39.5	Utility destroyed (57× increase)
10^{-2}	0.054	0.77	+12% loss
10^{-1}	0.083	0.70	+1.5% loss
≥ 1	0.91+	0.69	Utility preserved, detection intact

Pareto frontier: Suppressing L to chance level requires $\geq 12\%$ eval-loss degradation.

Known Failure Modes

1. **Non-residual architectures:** PlainNets show no signal (3% pair accuracy vs 100% for ResNet)
2. **Degenerate residual blocks:** Small-init ($\sigma=0.02$) on toy tasks where blocks collapse to near-zero contribution — fingerprint never develops (13% accuracy), but eval loss is excellent (0.004)
3. **Architectural scope:** Method requires identifiable residual-branch structure. Excludes:
 - Plain feedforward networks
 - RNNs
 - Architectures with non-linear gating without linear branch product

Protocol Failure Modes

Cannot provide verdict when: 1. Architectures differ 2. Perturbation destroys utility (e.g., 2-bit quantization) 3. Both models descend from unavailable common ancestor 4. Adversarial evasion specifically targets the fingerprint

What the Method Does NOT Do

- Does NOT distinguish two independently trained networks of same architecture (both satisfy $W_{out} W_{in} \approx -\epsilon I + E$)
- Does NOT detect training-data overlap or knowledge transfer
- Does NOT work without white-box access

Training Quality Assurance (Early Warning)

Condition	Final Pair Acc	Epoch 10 Acc	Neg Trace	Status
Healthy baseline	100%	8%	100%	PASS
LR too high (10^{-2})	100%	100%	100%	PASS
LR too low (10^{-6})	6%	6%	44%	FAIL
No skip (PlainNet)	3%	6%	47%	FAIL
High weight decay	24%	3%	0%	FAIL
Small init ($\sigma=0.02$)	22%	32%	31%	FAIL

Early warning protocol: - Pair accuracy $< 50\%$ at epoch 10 $\square$ WARNING - Negative trace $< 50\%$ at convergence $\square$ Blocks haven't achieved dynamical isometry

Correctly flags **4/5 pathological conditions**.

Summary Table

RQ	Question	Answer	Key Metric
RQ1	Emerges from training?	Yes — gradient coupling, not task learning or dynamical isometry	Shuffled labels: $s=5.88$ (72% stronger than normal $s=3.42$); $r(s,$
RQ2	Generalizes?	Yes — 12 architectures, 5 families	91–100% pair accuracy; GPT-2 s scales as $d^{0.87}$ (faster than $\sqrt{d}$)
RQ3	Discriminates?	Yes — AUROC 1.000 (CP TPR ≥ 0.961 , FPR ≤ 0.035 on $n=75/84$)	TPR 100% @ 1% FPR; 172× gap on ResNet-18 CIFAR benchmark
RQ4	Survives?	Yes — FT, quant, prune, LoRA	100% detection
RQ5	Resists/Fails correctly?	Yes — $\geq 12\%$ loss to suppress	Symmetries: $L = 1.0$

Mechanistic Understanding (Confirmed)

Hypothesis 1: Dynamical Isometry

Claim: Training makes Jacobians near-orthogonal, creating the fingerprint.

Status: ☐ **Falsified**

Evidence: - MLP: Jacobian orthogonality improves marginally (δ_J^{norm} : 0.100 $\square$ 0.089, -11%) - GPT-2: Jacobian orthogonality gets **worse** (δ_J^{norm} : 0.025 $\square$ 0.297, +12×) - Fingerprint works perfectly in both cases

Hypothesis 2: Gradient Coupling

Claim: W_{in} and W_{out} co-evolve under backprop because they receive correlated gradient signals, causing $M = W_{\text{out}} @ W_{\text{in}}$ to develop diagonal structure.

Status: ☐ **Confirmed**

Evidence from Section 9 experiments:

Experiment	Result	Implication
Part A	$r(g_{\text{diag}}, s) = -0.12$	Gradient product G is NOT diagonal; mechanism is subtler
Part B	Shuffling reduces s by 68%	Gradient coupling is necessary
Part C	Synthetic injection creates $s=7.6$	Diagonal structure is sufficient

The mechanism: Correlated gradient *directions* (not shapes) between W_{in} and W_{out} accumulate into diagonal weight structure over many updates. The same loss signal reaching both matrices causes them to co-evolve in a coordinated way that makes their product M diagonal-dominant with negative trace.

Pending Experiments (Issue #43)

The following items from the ablation issue require GPU/cluster compute we do not currently have and are queued for the next iteration. Each item lists what we would measure, the expected delta to current results, and a rough resource estimate. Scripts that can be drafted without running are linked.

Compute-bound (defer until cluster access)

Item	Owner ask	What we'd measure	Expected delta	Resource estimate	Script
MoE / Mixtral-8x7B fingerprint (RQ2)	@singh96aman	Per-expert and aggregate s , δ_J^{norm} ; pair accuracy across experts	Tests whether experts inherit the fingerprint independently or share it via the gate	~80 GB GPU, 1–2 hrs inference	experiments/scripts/rq2_moe
LLaMA-2-13B fingerprint (RQ2)	@singh96aman	Mean s , δ_J^{norm} , pair accuracy; extend GPT-2 $d^{0.87}$ scaling fit	Confirms or refines the scaling exponent at $d = 5120$	~30 GB GPU, < 1 hr inference	experiments/scripts/rq2_llama

Run Commands (SageMaker) LLaMA-2-13B (ml.g5.4xlarge or ml.p3.2xlarge — 24GB GPU):

```
# Option 1: Using transformer_family_pairing.py (full analysis with random baseline)
python experiments/scripts/transformer_family_pairing.py --model llama2-13b --seeds-random 3
```

```
# Option 2: Using rq2_llama_scaling.py (scaling analysis with GPT-2 comparison)
python experiments/scripts/rq2_llama_scaling.py --include-gpt2
```

Mixtral-8x7B MoE (ml.g5.12xlarge or ml.p4d.24xlarge — 96GB+ GPU):

```
# Full MoE analysis with per-expert fingerprints
python experiments/scripts/rq2_moe_fingerprint.py --model mixtral-8x7b
```

```
# Memory-efficient safetensors extraction (if model pre-downloaded)
python experiments/scripts/rq2_moe_fingerprint.py --model mixtral-8x7b --use-safetensors
```

Local-runnable (script staged; pending compute time)

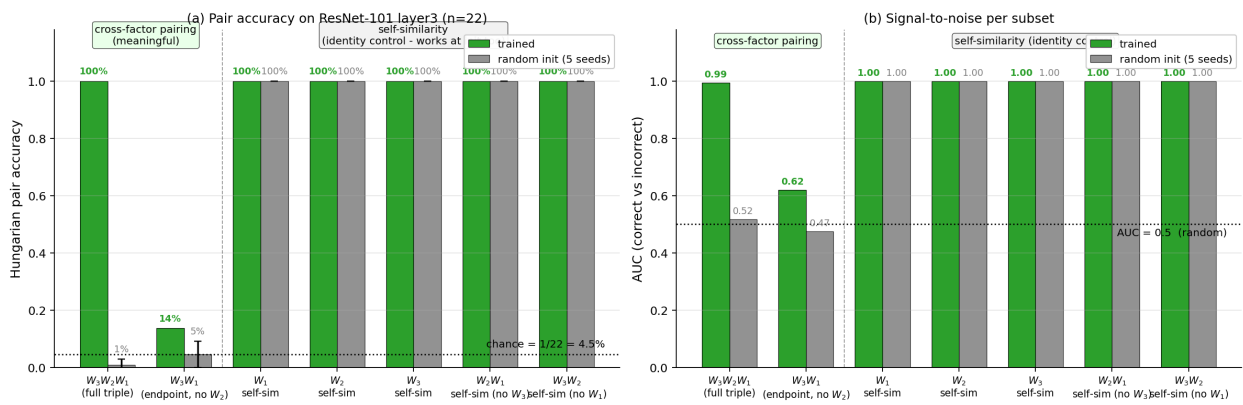
Item	Script	What we'd measure	Why it matters
Optimizer ablation (Issue #43, item 4)	experiments/scripts/rq1_Fingerprint_ablation.py	Final pair accuracy, emergence epoch, eval loss for SGD, SGD+momentum, Adam, AdamW, RMSprop on 48-block MLP	Tests whether the gradient-coupling mechanism is Adam-specific or general to any first-order optimizer. If SGD also produces the fingerprint, the mechanism is fundamental; if only momentum-using optimizers do, the mechanism depends on EMAs of gradients.
Cleaner $d^{0.87}$ derivation	(drafting)	Theoretical justification for the empirical scaling exponent	Replaces the falsified $\sqrt{d}$ claim with a defensible analytical result.
Per-kind $n \geq 30$ lineage benchmark	(drafting)	Per-kind TPR / TNR with CP CIs that have lower bounds > 0.90	Tightens the AUROC = 1.000 result on per-kind basis.

All Figures Gallery

Core Methodology

Deep Dives

Bottleneck factorization: endpoint W_3W_1 fails, full branch product $W_3W_2W_1$ succeeds



ViT-B/16 attention: which heads carry the pairing fingerprint?

